Bitcoin: Independence Reimagined

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Foreword by Hodlonaut



21M

Lemniscate Media

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I dedicate this book to all of those who helped me with my last book, “Sovereignty through mathematics”

Translators, podcasters, proof-readers, artists, tweeters, box makers, site owners and interviewers etc... The Bitcoin community is the least greedy, most helpful and most honest group of people I’ve ever encountered.

Thank you!

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Foreword

Being a space cat, I have a rather independent mind. I like to think for myself, I like to question things that don't make sense and I like to speak what I believe to be true. I consider my opinions mostly free from outside control, not subject to the authority of others. This is likely part of the reason why I fell so instantly and hopelessly down the rabbit hole after discovering Bitcoin. I first saw Bitcoin mentioned in some tech magazine article back in 2013. It was like being out fishing for information, like every other day, and suddenly catching an idea so big, so novel and alien, it changes everything. The catch of a lifetime. For many people, reeling in such an idea would lead to them frowning and throwing it right back into the ocean. It didn't fit into the existing system, and more importantly, no authority was there to validate it for them. Mainstream media and different kinds of experts went out of their way to ridicule and demonize the idea. Thus, for most, it didn't make any sense to keep it or have a closer look. The idea we call Bitcoin is so big and so disruptive that it doesn't ask to fit into the system. It aspires to change the system itself. Entertaining such an idea requires a certain amount of independent thinking.

Swiss psychologist Jean Piaget had an interesting take on learning. He described how, when we encounter a new idea, we settle on one of two fundamental responses.

We either try to fit the new idea into what we already know, similar to filling it into an existing box. This is called *assimilation*. The framework to understand the idea is already in place. It's easy and doesn't require us to question or change any of our pre-held beliefs. This approach works with most new information we encounter in our daily lives. But sometimes, the idea doesn't fit into any of the boxes we have. This means we must reshape or expand our boxes for the idea to make sense. This is called *accommodation* and is a much more substantial and demanding process. This is why an idea as great and novel as Bitcoin can be at risk of being thrown away entirely or assimilated/shoehorned into a box it doesn't fit in. Bitcoin is definitely an idea that requires accommodation to be meaningfully understood. I would argue that the Bitcoin journey is a journey of almost continuous accommodation. It drives us to question the validity of one knowledge box after another. As the title of this book suggests, it even enables us to reimagine core concepts of life like independence.

I was first exposed to the Bitcoin writings of Knut Svanholm when I listened to Guy Swann read his first book, *Sovereignty Through Mathematics*, on The

Cryptoconomy podcast[1] . It is an excellent book, providing a solid framework to understand core concepts like money, inflation, and scarcity and why Bitcoin is such a paradigm-shifting idea. With *Independence Reimagined*, Knut is giving us a lot of very valuable insights that will further help us in our efforts to accommodate this incredible idea called Bitcoin. He cogently describes the old boxes we have been operating within and how Bitcoin encourages us to challenge and reshape them. He improves on the already existing mental scaffolding for us to better understand the enormous beast of an idea that is Bitcoin. How we can grasp its implications, its potential for change and how it can be used as a tool for us to improve our independence in a time where it is sorely needed.

Hodlonaut, April 2020

1. Now rebranded to the Bitcoin Audible podcast.

Disclaimer

The viewpoints expressed in this book (as in my other books) are primarily meant to provoke thought. I hope that it will inspire you to change your perspective. Not necessarily into my perspective, but a change nonetheless. This holds true for my previous book, *Bitcoin: Sovereignty Through Mathematics*, as well. Neither I nor anyone else can know with certainty whether or not the world will be a better place after the implementation of our proposed solutions to the world's problems. We cannot know if the ideas of the Enlightenment came to be because of a well-organized Judeo-Christian societal base layer or if the religious dogmas of the past have just been hindering scientific progress by calling it blasphemy for centuries. Neither can we know if technological progress is a self-destructive process or not. Our history can't be altered, and our future is unknown. We're all chained to the unforgiving arrow of time. All we can do is speculate and educate ourselves. Regardless of what opinions we hold and how those opinions influence society as a whole, we can still study what tools we as individuals have at our disposal to regain a little control of our lives — if we're prepared to think outside of our boxes, that is. If we're really willing

to take responsibility for ourselves and take matters into our own hands, all the tools needed are at our disposal. All of them are cheap. A Raspberry Pi and an Internet connection are quite affordable for most of the world's population. What are no-coiners waiting for?

Finally, I hope you'll enjoy consuming this thought-nugget buffet as much as I enjoyed cooking it.

Knut Svanholm, January 2020

Fiction and Reality

Humans are unlike other animals. Our ability to cooperate on massive scales certainly seems to be a skill other apes or mammals lack. What is the unique mental tool that makes us capable of doing this? Our closest evolutionary relatives, chimps, baboons, and other monkeys, show similarities to humans when it comes to forming hierarchical societies, but none of their packs or tribes ever grow larger than a few hundred individuals. Why is that? It has something to do with our collective imagination. We've observed chimpanzees constructing altars for worship, dancing rain dances, and grooming their dead, but they are yet to be seen pilgriming by the millions from all continents to kiss a wall. They trade fruit and present offerings to their patriarchs and matriarchs, but they aren't yet trading government bonds for fiat currency. We humans, on the other hand, have conquered the world with our ability to rally under a common flag. World religions, spiritual movements, nation-states, and even multinational corporations have provided human beings with the motivational tools necessary to take control of the earth and dominate all other forms of life, even though the concept of a nation, a religion, or a company

are all just social constructs of human imagination at their core. We humans tend to perceive our collective imaginations to be just as real as the ground we walk on, the foods we eat, or the air we breathe.

This ability to lie to ourselves at scale is arguably our biggest weakness as well as our greatest strength. We've somehow managed to conquer the world while concurrently setting a trap for ourselves that we have already fallen into. Blindly following our leaders has rendered us under-equipped to detect when we're being used. *We've been domesticated.* If the idea of domestication bothers you, you need to free yourself from your leash. This might be harder than you think and you'll soon realize that you're tied to more than one of them. So is everyone else. Not listening to religious lunatics might be easy, but not buying into the idea of the nation-state, or even democracy, is a stretch too far for most people. When you really think about it, though, borders, laws, institutions, and ultimately, money are just as intangible as Zeus, Thor, or Sauron. Only our common agreement of their existence makes them exist. Unlike trees, cows, or garbanzo beans, they could not exist outside of our minds. The fictitious nature of money, in particular, is vastly misunderstood. "Permit me to issue and control the money of a nation, and I care not who makes its laws," Mayer Amschel Rothschild allegedly said. At the very top of each nation's food chain sits the entity controlling our money, the lifeblood of every society.

So what can you do about this? And why would you want to do anything about it in the first place? You may have a perfectly fine life right now, with a steady income and many career opportunities. So why bother questioning everything? If we look at certain statistics presented by the likes of Hans Rosling and Steven Pinker, it becomes abundantly clear that the denizens of Mother Earth are doing better and better every year. While the claims of Rosling and Pinker may be somewhat truthful from certain points of view, they fail to address the long-term problems inherent in our fiat currency systems. They fail to address that if we continue business as usual, we're all doomed, not because of climate change, nuclear war, or any other red herring, but because of the horrifying Orwellian dystopia that will be the inevitable outcome of our current pandemic monetary cancer. You may not be aware of it, but your country's currency is on the exact same path as that of the Venezuelan Bolivar or the Turkish Lira. The only thing that differentiates them is the pace of the decline. The compound interest of a two percent inflation per year currency will sooner or later follow the same hockey-stick-to-hell curve as any other hyperinflating monopoly paper money. Not only this, but as physical cash becomes more rare, your financial affairs get more surveilled. Your banker overlords are monitoring you, mapping you out, and categorizing you according to your financial behaviors. To them, you're just a piece on a board game that they wrote the rules for. As long as you stay in the conventional systems, there's nothing you can do about it. Your vote won't matter. You're not in control of your destiny — not by a long shot.

But you could be if you found another way. If you found a way to store the fruits of your labor in a sort of monetary freezer that made sure that the number representing the work carried out was representing the same amount of work or more when you defrosted it again. A cold storage of wealth. A piece of a specific pie. A piece of Bitcoin.

Is money real or just another human collective hallucination? Well, a gold coin is real in the sense that it exists and that it's made up of gold atoms. Its value, on the other hand, isn't real. It can't be. Value is, contrary to what most people believe, an entirely subjective thing. There is no inherent value to any object. Value is a human creation that we imbue to objects or services. What makes a gold coin valuable to a person is that person's belief that gold will stay valuable to a lot of other people. Value is determined by the importance an acting individual places on a good for the achievement of his desired ends. It is pure speculation, and so is everything else in economics. If you buy a bottle of water, you do so because you think that its contents will satisfy one of your basic needs (probably thirst) at some point in the future, namely when you open the bottle to drink it. What makes a gold coin valuable to a great number of people is the perceived scarcity of the metal. Making jewelry out of gold doesn't give gold its value. Jewelry is made out of gold because jewelers believe they can profit from the metal's perceived value. Value, while being subjective, can still be said to be derived from supply and demand. We think we know that gold is a rare metal, but is it? No single human can ever truly know, but there's enough

verifiable data for us to believe that the global supply of gold is somewhat limited. It is this believed scarcity of the supply of a good in conjunction with the belief that there will be a future demand for it that makes humans attach a value to a good or commodity. Because dollar bills were once redeemable for one dollar's worth of gold at a bank, people still think that dollars are connected to gold somehow or, at the very least, that they're somewhat scarce. Spoiler alert — they aren't. Bitcoin, on the other hand, is verifiably scarce by anyone running a full node in the Bitcoin network. This feature alone makes Bitcoin a far better form of money than all of its predecessors.

Money, laws, nation-states, religions, and other constructs of the human imagination are not entirely subjective beliefs, nor are they objective facts. They're not entirely subjective because their existence is not limited to the brain of a single individual. A better, more suitable description would be that they're *inter-subjective*, meaning that they exist in the collective consciousness of large groups of individuals. Because of this, ideas like that of money or that of an almighty God, for that matter, do not die because of the death of a single believer in that particular idea. Ideas live on until they're destroyed by force or until a better or more saleable idea comes along, rendering the old idea obsolete. Whatever spirit gods the natives of Cuba believed in, for instance, disappeared when the European conquerors murdered every tribe that inhabited the island. The idea of alchemy, on the other hand, became obsolete with the rise of chemistry, which rendered the idea of alchemy somewhat

ridiculous in hindsight. This very text is only able to convey its message because of the inter-subjective idea of the English language, without which the words written would be little more than the undecipherable scribblings of a madman. Our current dominant monetary systems are inter-subjective ideas based around the "full faith and credit of" nation X's central bank. They are very powerful short-term and can help infrastructure reforms at a fast pace, but they do so at the expense of future generations who have massive debts, authoritarian regimes, hyperinflation, and possibly civil wars to look forward to because of these flawed systems. Bitcoin doesn't need an inter-subjective belief other than that miners will continue to act in their own best self-interest, therefore its promise of a finite supply is more of an objective fact than an inter-subjective fiction. Its functionality and finite supply are at all times verifiable by the network's users. The only other inter-subjective belief needed in order for the system to thrive is the belief that people will continue to discover its superior monetary properties — properties that cannot be reproduced since Bitcoin is so much more than the sum of its parts. Its code, its developers, its noderunners, its surrounding software, and, first and foremost, its history makes it a truly unique beast.

If the system continues to function as it has for the last eleven years, hyperbitcoinization will happen. It is inevitable. The only uncertainty about the transition to a Bitcoin-denominated world is the timeframe. We cannot know how long it will take for the fiat currencies to

die, but it probably won't happen within the next couple of decades. After all, the idea that Bitcoin challenges is arguably the most successful inter-subjective idea that ever existed — the idea of a global reserve fiat currency. It is an idea that cements the planet's current power structures and hierarchies and keeps populations in check. It won't be easy to persuade people that everything they thought they knew about money, banks, jobs, income, taxes, laws, and even democracy was based on a lie and that this lie has now been exposed. In truth, just about the only thing that can persuade people to switch to Bitcoin is the number that goes up over time — the price of Bitcoin. A limited supply doesn't automatically mean unlimited demand, and it is important to remember that Bitcoin is still being issued at a faster rate than gold and most fiat currencies right now. At the time of writing, the halving from 12.5 to 6.25 new Bitcoin mined per block is about five months away. Don't forget that we're still in Bitcoin's early history and that a lot of volatility is still to be expected for any foreseeable future.

Cooperation and Coercion

"*Workers of the world, unite!*" Marx and Engels commanded in the Communist Manifesto. "*You have nothing to lose but your chains!*" The idea was to get the workers of all nations to cooperate in order to overthrow the Bourgeoisie and revolutionize the world. Turns out they had quite a lot to lose and that they didn't necessarily lose their chains in the process, either.

Involuntary cooperation is not an effective economic process. It steals time and effort from those involved. When a person is disallowed to pursue his own goals, he is also stripped of the will to cooperate with others to reach these goals and realize his dreams. A person has but two tools at his disposal to gain the cooperation of others: one is speech, and the other is violence. Money is the linguistic tool he uses to express how much he values the other person's efforts. It's an intersubjective agreement that works because both parties believe it to be true. They believe it to be true because they believe that the money exchanged will hold its inter-subjective value long enough for the receiving party to exchange it for another good or service. When the money used is inflationary and therefore expected

to lose value over time, people are incentivized to use it earlier rather than later. In other words, it makes people think that having a high time preference is something to strive for. Keynesian economic theory, which endorses inflation, makes people believe that fast, unconsidered transactions are what drive an economy forward rather than slow, thoughtful ones. Governments love these theories since they keep citizens indebted and submissive.

Austrian economic theory (or *reality*), on the other hand, teaches us that in order for a civilization to thrive long term, its population needs to be able to make informed investment decisions long term because this process is the very foundation of civilization. The voluntary cooperation of two or more humans is the *civilized* way of doing things, hence the very word *civilization*. In this sense, the private sector is much more publicly owned than the so-called public one, which is basically made up of redistributed stolen goods. Public spending is like a game of poker without bets. When a person has nothing to lose from bad decision-making, bad decisions will be made. When a person is held accountable for his or her actions (when that person has *skin in the game*), he will make more rational decisions and fewer irrational ones. This is not to say that every private investment is sensible; it is just that having something to lose is a key underlying requisite for sound decision-making.

Taxation, and even more so inflation, are inter-subjective fantasies with such worldwide acceptance that it's hard to imagine a world without them. However, neither one of them is rooted in voluntary cooperation. In order for a government or a ruler to impose taxes or inflation on its subjects, it needs to coerce them into submission first. This cannot be done without a threat of brute force backing the legitimacy of proposed policies, a fact easily forgotten as taxes and inflation exist practically everywhere on the planet. There are, however, other ways of financing large cooperative ventures, they just haven't been tried at nationwide scales yet. Crowdfunding, for instance. The Internet has given us a plethora of financing tools that have the potential to change everything.

Bitcoin is a difficult concept to wrap your head around because it requires you to question and re-think so many things that you've mistaken for absolute truths all your life. Democracy, for instance. We've all been told in school that democracy is the best way there is to run a country because it revolves around the will of the people. The schools themselves are completely authoritarian and run in a top-down fashion, though — institutions that all democratic countries force their young citizens to spend at least nine years of their lives in. We find a governing body at the very top that decides on the curriculum and ranking of establishments. On one level underneath, we find an unelected bureaucrat (the principal or headmaster) who, in turn, governs the next layer of authoritarian rule and law-upholders

(the teachers) who are in charge of controlling the serfs. A poor mass of sorry kids who are force-fed nationalistic and biased information across all subjects (even including math, sometimes) and punished if ever found challenging the order. No questions in class, just sit down, shut up and do your work.

We can vote for this or that politician, but we can't vote ourselves out of democracy itself. It's not voluntary. Democracy is marketed as the only legitimate way of governing a country. In almost every school, it's compared to nothing but the systems of power that preceded it, namely feudalism, religious rule, plain old dictatorship, or despotism. These systems often involved some form of hereditary rulership. When democracy came along, people felt empowered since they had a say in the decisions made, but the coercion didn't disappear. Taxes didn't go away, and the right to print money certainly wasn't given to the people. In the early stages of the democratic process, whereupon many of the big companies that entire nations rely on for their prosperity today were founded, taxes were generally low, and there weren't that many institutions in place. For example, in my own country of birth, Sweden, taxes were below 20 percent between 1850 and 1950. During this period, income per capita in the country increased eightfold. The average age of a big Swedish company is 94 years, which means that most of them were founded during this era. In 2004, when Swedish taxes had more than tripled, and social security programs had been in place in the country since at least the seventies, sixteen percent

of the government's budget went to sickness benefits for Sweden's objectively rather healthy population. A large public sector requires a lot of administration, and administration spawns bureaucracy. Institutions and bureaucrats, once given a right to exist, soon become experts at defending their own necessity no matter how absurd their initial supposed function may be.

In the long run, the coercive methods of big government and large corporations become very expensive to maintain. To keep up with the rising costs of keeping the ducks in line, we get phenomena such as negative interest rates, bank bailouts, and quantitative easing. There is a limit to how much a government can earn from taxation, as shown by the *Laffer Curve*, but not all governments are aware of this, and few see their economic downfall coming before they're already in an unsustainable position. No matter how bad these behemoths become at their jobs it is always the little guy that will have to pay for the whole charade. Bitcoin fixes this for you by refusing to take part in any of it.

We're all coerced into some behaviors. All of us live in a certain jurisdiction with its own unique set of laws and regulations. Of course, many of these laws, if not most of them, are designed to protect us from violent coercion in one way or another. However, the very notion of a law book governing our actions alters how we interact with each other in many subtle ways. If a person knows about a law and its repercussions, he is unlikely to break it intentionally, even if the supposed "crime" is victimless

and hurts no one. If he has a lot to gain from breaking the law and believes that he can do so without too much risk, he will break it, but only if his personal sense of morals permits him to do so. Because of these relatively high thresholds to a person's decision-making processes that the existence of a law infers, most people just try to live their lives as if they agree with every policy and law their government chooses to impose on them.

Why Collectivists Win

People, in general, want to be free to do whatever they want with their lives. Even left-wingers seem to have their personal freedoms as an end goal, even though personal freedom means something else to them than it does to those actually capable of supporting themselves. A lot of people see welfare as an opportunity to live their lives as they want, as starving artists, hermits, bums, or junkies. If you're a nihilist, depressed, or just not very materialistic, you might think that living off of social welfare for the rest of your life is actually a viable option. It's not really immoral to crave as little as possible, is it? Nor does it really deprive you of your freedoms, does it? Well, ask a Venezuelan what he thinks about such claims. The problem with giving away more and more power to politicians is that you become defenseless when the system implodes, which it inevitably does one day. Once a draconian law, a bureaucratic institution, or an infringement of your right to freedom of thought is in place, it is very hard to remove it again without an outright revolution. A revolution is always violent and hardly guarantees any better shot at independence than what you had before it, though.

People keep voting for more and more government interference while they keep complaining about the government interfering in their lives too much. Why? Let's examine what attributes define a specific voter. An individualist voter tends to care less about what others think about his political views than his collectivist counterpart. A person who cares about what others think of his opinions is, by definition, less individualistic than a person who doesn't. What this inevitably leads to in a democracy is that individualists will tend to vote for a greater number of, hence smaller, parties than collectivists will. A collectivist naturally tends to care more about belonging to a group, per se, and less about what that group actually wants.

Individualists, on the other hand, are less agreeable than collectivists because whether they share their opinions with others or not matters far less to them than it does to the collectivists. In other words, collectivists form larger groups more easily than individualists. This is one of the most subtle yet most dangerous pitfalls of democracy itself. In the long run, every government, every state institution, and every politician is bound to become corrupt because the forces that spawned them were more concerned with tribalism than individual rights. Psychopaths thrive in such environments. There's just no way of telling if a seemingly polite and friendly political leader is really a cold-hearted, calculating monster underneath his facade.

It's depressingly easy to get a political or even a religious belief to be considered an intersubjective truth. You can count on an enormous amount of pre-existing credulity among illiterate, frightened, ill-educated populations. In Sweden, we were repeatedly told about every human's equal value in school. As I recall, none of the kids ever questioned this. No one ever mentioned that value might not even be something you can measure because of its subjective nature. Those asking the really hard questions are seldom rewarded for doing so by the group, no matter what type of group we're talking about. The larger the group, the more intolerant it becomes of the skeptic, or as George Orwell put it: *"The further a society drifts from the truth, the more it will hate those that speak it."*

Political correctness is a perfect example of this phenomenon. The words "climate denier" and "islamophobe" are examples of how even our language has changed in order for the mainstream to quell undesired opinions. Most people labeled with these epithets are merely critical of the way the world is presented to them by mainstream media and not in denial or phobic by any stretch of the imagination. The word "denier" implies that there is an absolute truth somewhere that ought not to be questioned by serious thinkers. What this truth actually consists of is unclear, though, since those who just refuse to buy into the idea that politicians can change the weather are labeled in the same way as those who actually question scientific data. Those who criticize religious dogma are labeled as

phobic towards the people who identify as belonging to that particular religious group.

A lot can be said about the late Christopher Hitchens and his criticism of religion, but you can hardly say that he was "phobic" of Muslims. On the contrary, he confronted religious leaders without fear all the time. On the other side of the political aisle, the word "pro-life" comes to mind, which implies that all those in favor of liberal abortion laws are murderers. As discussed earlier, speech is the only tool we humans have, apart from violence, to resolve our conflicts, and this type of political censorship of our language always seems to stem from political or religious groupthink. Large flocks of people mimicking each other's behavior can be a very scary thing, as those who have ever found themselves in the midst of a riot can attest to. The most hideous atrocities ever committed against humans have all been made possible by armies of collectivists following orders and not questioning the morality of their actions.

If you're honestly frightened about whatever political direction your nation seems to be headed in, you have two options. You can either stay in the illusion, believe that your vote will account for something, and hope that everything will magically sort itself out one day, or you can choose to try to regain a little control over your own destiny. Just like in the movie *The Matrix* from 1999, you have a choice. As the character Morpheus says:

> *"This is your last chance. After this, there is no turning back. You take the blue pill — the story ends, you wake up in your bed and believe whatever you want to believe. You take the red pill — you stay in Wonderland, and I show you how deep the rabbit hole goes. Remember: all I'm offering is the truth. Nothing more."*

Bitcoin offers you a similar choice. To properly understand Bitcoin and why you need it, you must first see the world as it truly is, and you might not like what you're about to find out. A curious mind just can't help itself, though, now, can it?

One thing a curious mind ought to be capable of doing is memorizing 24 words. Since memorizing 24 words enables you to cross any border in the world with any amount of money in your head, this is nothing short of a superpower to a Venezuelan refugee or a Chinese dissident. Information has always been a valuable thing, but since the inception of Bitcoin, information can literally hold inter-subjective value. Of course, value is still a subjective thing, but the inter-subjectivity of something we can define as money can be a very useful tool. You don't have to remember all of the 24 words either — just remembering four of them should provide you with enough entropy to create a time slot long enough to travel from one jurisdiction to another with any amount of wealth in your memory, enabling you to recover the whole seed phrase, and with it, access to your Bitcoin, if you recombine them with the 20 other words

which you had already submitted to the destination before. This is just one of the ways in which Bitcoin will inevitably make the nation-state obsolete.

Can a nation hold Bitcoin? Can a bank? It is often speculated that nations, banks, and institutional investors are going to start stockpiling Bitcoin, but Bitcoin is not just another currency or asset. They cannot be owned by a non-existing inter-subjective entity in the way traditional money can. It is even questionable whether they can be owned at all since all that owning Bitcoin means is that the owner has access to them by holding the private keys to a certain Bitcoin address. A nation-state is made up of people. So is a bank, and so is any other man-made institution. Someone in the allegedly Bitcoin-owning institution would have to be entrusted with the private keys. Of course, there are multi-sig setups, and you could picture a nation holding Bitcoin in a way similar to that in which nuclear missiles are handled in Hollywood movies — a room with multiple literal keys for a bunch of uniform-wearing minions to turn simultaneously when a big red telephone rings and the President commands them to do so. The difficulties an institution would have in holding Bitcoin illuminates a bigger problem — even in our current system, there is ultimately a need to trust a single individual at the very end of each ownership chain. Institutions aren't real. Banks, and even more so central banks, are only possible because of people's belief in the validity of what used to be a receipt for a specific amount of gold deposited at that bank — the banknote.

Even today, when people ought to know that no currency in the world is redeemable for gold any longer, they act as if they were and take their supposed value for granted. With Bitcoin, this is not possible because Bitcoin is not a receipt for anything; Bitcoin *is* the scarce asset its private key represents. This turns everything we thought we knew about money on its head because it exposes the biggest lie we've ever fallen for as a species. The liars in power on top of the food chain of every human hierarchy are losing their ability to spellbind their subjects more and more for every day that passes. The emperor is naked, and Satoshi Nakamoto has pointed it out. Everyone will realize this eventually because it's right in front of their eyes. There is no such thing as a society, and there never was. It was just you, me, and the others all along. There is no we. There is no ours. There are only individuals with different amounts of influence on other individuals. Only an individual can make a decision. Only an individual can claim ownership.

Collectivists win because of their ignorance. Because of their willingness to be used. Because they fail to realize that it's all in their heads. All of it.

A Tale of Two Richards

In the 1680s, on the Emerald Isle of Ireland, a boy named Richard Cantillon was born. He grew up to be a successful merchant and banker and wrote books on economics and monetary theory. He's best known for noticing the economic phenomenon named after him, *the Cantillon Effect*, which describes the asymmetric economic effects of inflation.

In a *fractional reserve* banking system, the financial system that the majority of the Earth's nation-states currently operate under, new money is created out of thin air when a bank decides to grant a business owner a loan. At a specified date in the future, this loan will have to be paid back to the bank with interest. The amount of money needed to make up for this interest is never in circulation at any specific point in time; therefore, there is a constant need for new loans to be granted. In other words, a need for new money is always present in the system. What Cantillon noticed, way ahead of his time, is that the closer a person is to the money being issued, the less of a negative impact the inevitable inflation following the newly expanded total money supply will have on that person. Inflation will, therefore, always

benefit those responsible for the money printing and their friends first. This effect is not limited to money but applies to anything with an increasing supply. There's inherent inertia in the price discovery mechanisms of all markets. The effects an increased supply of a good will have on its price are not felt directly but gradually as the relationship between supply and demand is altered. If you can produce more of something at a lower cost than your competitors, you reap the benefit of charging a higher price before the market discovers that there's an oversupply of that something.

One example where the Cantillon Effect can be observed in a commodity other than money is in the value of something as odd as autographed restaurant bills. Sometime in the mid-twentieth century, the by-then world-famous painter Salvador Dali allegedly signed the wreckage of a car he had crashed into in France, instantly making it more valuable than it was before the crash. He had, by then, made such a name for himself that putting his signature on stuff instantly increased the value of whatever thing he chose to honor with his autograph. He was not the only famous painter who did this, though. Several others had figured out that they could pay their restaurant bills by simply signing them. By this act, they magically transformed small pieces of paper into "art." These autographed bills were then traded among the restaurant's staff and elsewhere. At first, they could be traded at very high prices, but as the practice became more common, their value naturally diminished. In other words, they became inflationary and therefore decreased

in value. The closer you lived to a celebrity painter, the more aware you were of how happy they were putting their names on practically anything, and thus, the more aware you became of how abundant these signed bills really were. In knowing more about the supply of this "good," you could also predict that they would lose value eventually — not at all unlike how a pyramid scheme works. This is exactly what Cantillon observed. Anything that can be marketed as more scarce than it actually is can be sold at a higher price as long as market participants haven't yet noticed its abundance. This always benefits you more the closer you are to the source.

In 1971, a more famous Richard did something to the world economy that all of his predecessors had resisted the temptation of doing: he suspended the convertibility of the US dollar to gold. This man's full name was Richard Milhous Nixon, and he was the 37th President of the United States of America. The US dollar had, until then, been redeemable for gold at a fixed price of 35 dollars per ounce. This price had been the true inter-subjective value of one American dollar since 1944, when representatives from 44 countries met in Bretton Woods, New Hampshire, to sign an agreement that effectively turned the US dollar into the world's reserve currency. After the Bretton Woods agreement, every other currency could be converted into dollars at an almost entirely fixed rate, and a dollar was considered "as good as gold" by most of the other participating states. It turned out that it wasn't nearly "as good as gold,"

though, as inflation ramped up in the US during the early seventies. Milton Friedman described the inflationary process and laid blame on the printing presses of the Federal Reserve in his 1980 TV series *Free To Choose*:

> *"The reason we have inflation in the United States, or for that matter anywhere in the world, is because these pieces of paper and their accompanying book entry for their counterparts in other nations, are growing more rapidly than the quantity of goods and services produced. The truth is — inflation is made in one place and one place only, here in Washington."*

The Federal Reserve had been doing what banks always do — it had been giving out way more receipts for gold (dollar bills) than they could pay for with the gold they actually had in their vaults. When foreign banks became suspicious and wanted to reclaim their gold, President Nixon simply changed the rules and made this impossible. By the time he did, the dollar was already established as the world's reserve currency, and there was nothing the other countries could do to change this. An ounce of gold is right now, in December 2019, worth 1 464 US dollars. In other words, a dollar's worth of gold today would be redeemable for less than two and one-half cents back in 1971. Banknotes and coins are rare these days, though, and most of the money in the world exists solely in digital form. Today, money in digital form is easier to conjure up out of thin air than banknotes were in the seventies. The superpowers of the world tirelessly

try to outdo one another in this department, now more than ever. Negative interest rates are a testament to this, and the predictions of the eldest of the two Richards described in this chapter continuously prove true as American and Chinese megacorporations take over the world at an ever-increasing rate.

Some say that Bitcoin is too experimental and unreliable. Who are we to blame if no one's in charge? Well, we're living in the time of the greatest unreliable monetary experiment ever conducted, and the person to blame is Richard Nixon. It's that simple. If you look at the statistics, much has changed since 1971. Between 1971 and 2017, the average productivity per hour of labor in the United States grew by 246%, while the compensation for those hours grew by only 115% during the same period. Wealth has been funneled into the pockets of the top one percent and out of the pockets of everyone else at an ever-increasing rate since Nixon's scheme began. National debts have risen. Housing costs have risen. Everyone's indebted, it seems. We all live in houses built by our parents and grandparents, but we *still* owe vast amounts of money to some bank for some reason. Most people just accept the way things are and think that there's nothing to do about it, but it is not the way the world is supposed to work. We've all been robbed, and we're still being robbed *right now*. The only way to stop the thievery is to quickly acquire something that is hard to confiscate and will hold its inter-subjective value for a long time.

All of the economic theories that rely on a constant inflation of the money supply have one thing in common. They're immoral. Whether they "work" or not isn't the point. If your theory needs to rely on thievery in order to get people to behave in certain ways, it is wrong. Period. A society that relies on the constant robbery of its subjects in its base layer is bound to be corrupted over time. How can anyone trust anyone else if pricing mechanisms are broken? Prices are information, and more than that, they're supposed to incentivize people to make sound, informed financial decisions so that resources aren't misallocated or wasted. As long as the world's currencies are inflationary, we can forget about equality, fairness, or environmentalism.

A corrupt system simply won't reward honesty. Without a moral foundation, every other socioeconomic issue becomes a mere charade. Mainstream media news outlets constantly focus on trivialities and make us forget the mass theft that created the power asymmetries in the first place. Have you ever asked yourself why TV scheduling is referred to as programming? No one in charge of money issuance has ever been able to resist the urge to enrich themselves at the expense of everyone else. The only way to protect yourself from the greed of the central bankers is to opt out. Ask yourself what ownership really is and which, if any, of your possessions are truly yours. Which of them can't be taken away from you because of a political decision or a banking crisis? Do you even have any items that fit these criteria?

The realization that every economy on Earth is a pyramid scheme can be very depressing. It forces you to think about how destructive your own behavior can be to those at the bottom of the pyramid. When you notice how unfair the whole system is, it's easy to become cynical and suspicious of everyone. Try to focus on what works instead and beware of conspiracy theories. Our cognitive biases skew our perceptions, and because of this, the "Don't trust, verify" approach is *always* the correct way of tackling a problem.

Don't guess, look it up! All of it!

Poverty and Prosperity

"The gaps are increasing!"

How many times have you heard this rhetoric? Whether it's the wealth gap between countries or the pay gap between the genders, the gap itself is always marketed as a vile thing and a societal problem that needs to be solved. We're led to believe that wealth cannot be acquired by any other means than by exploiting the poor in one way or another. The Uncle Scrooges of the "one percent" are somehow leeching off of the lifeblood of the less fortunate, using them for their evil capitalistic schemes. This rhetoric is completely backward, though, and merely serves as a tool for the real elite to keep all of us ducks in line.

First of all, economics is not a zero-sum game. John, the fisherman, can sell fish to Bob, the builder, and Bob can build John a fishing rod in order for him to increase his daily catch. By doing this, John can lower the price of fish, which is a net good (pun intended) for everyone in the economy (except the fish). In an economy with perfectly sound money, in other words, money with a

fixed supply, the value of the medium of exchange used in the economy doesn't go down but up, making everyone in that economy wealthier as the economy grows. We're so used to operating under inflationary currencies that only a select few of us have experienced this, though. Of course, we've all seen the prices of goods that have had exponentially decreasing production costs, such as electronics and computing power, go down, but in a sound money economy, *all* prices go down as overall productivity rises. If the amount of money is stable while the amount of goods we can produce increases, it will necessarily take less money to buy more goods.

Historically, only two phenomena have been able to actually decrease the wealth gap between people: natural disasters and war. It is unclear which of the two categories socialist policies, interventionism, and central banking fall under. In a well-oiled economy, wealth gaps will increase naturally since more productive people will always accumulate wealth at a higher pace than less productive ones. This phenomenon is a good thing and of benefit to everyone because in order to monetize his productivity, man needs to serve other people's needs. Every capitalist is a slave to his customers' wants and needs. Unfortunately, no one on Earth has ever been able to enjoy the full potential of a truly free global market because we've never had a functioning, sound money economy up and running properly. Why not? Because the temptation of money printing is simply too powerful for the human mind. No nation's central bank or government has ever resisted the urge to debase its national currency

for its own benefit. Bitcoin is a different beast altogether, and its issuance rate is beyond the reach of mere humans. It can't be tampered with, regardless of temptations.

In a sound money economy, and arguably in any economy, you can't fight poverty by fighting prosperity. In a sound money economy, everyone profits from economic growth. The road out of poverty inevitably leads to more and more prosperity. To set a limit to this or to interfere with the process will hinder a person from becoming wealthy, but it will, at the same time, hinder his potential customers from acquiring the tools they need to lift themselves out of poverty. In a truly free market sound money economy, wealth taxes are a net loss to everyone. The costlier a monetary good is to produce, the cheaper everything else becomes compared to it, and vice versa. The cheaper the money, the costlier everything else. As time goes by, Bitcoin becomes costlier and costlier to mine. It looks like Bitcoin will hold this property forever since the last coin will be mined around the year 2140. After that, minting new Bitcoin will be close to infinitely expensive. This means that everyone who holds any Bitcoin will always profit whenever the Bitcoin economy grows, just as everyone loses from holding fiat money.

Some claim that Bitcoin was designed for black markets and that without state control, money laundering and other economic crimes will be harder to fight against in an economy without government oversight. From an Austrian economics perspective, this

is absurd and just another proof of how effective the brainwashing machine of the interventionists has been. To a libertarian, there is no such thing as a black market since markets aren't intrinsically good or bad, and there's nothing morally wrong in trading one item for another. In well-oiled, functioning liberal democracies, the necessity for black markets is somewhat limited, but the black markets of Venezuela, Zimbabwe, or Cuba tell a different story. White markets might keep so-called dangerous goods off the streets, but they reinforce the status quo, let cartels have things their way, and form monopolies. We are yet to see the final result of the last century's worldwide interventionist policies, but while we wait, there's only one type of tamper-proof way to transact peer-to-peer besides barter. That ought to mean something.

Have you ever walked into a brick-and-mortar store and asked them where you could find something that they didn't have? The answer you can expect is one that recommends another, often competing store. Despite the fact that the first store has nothing to gain from you going to another store instead, this level of courtesy can be expected almost anywhere. Why is that? Shouldn't the evil mechanics of the free market system have thwarted such behavior in brick-and-mortar store employees by now? Quite the opposite! Markets evolve into friendly places because a seller's top priority is always the satisfaction of the customer. It has to be. If a customer gets irritated or angry because of the attitude of an employee of a certain store, he will cease to shop there.

An entrepreneur is always a slave to his customers' wants and needs. The flip side of this coin is that the customer is always the master of the entrepreneur, but only if he's in control of his wants and needs.

In order to be in control of one's wants and needs, one needs to be self-sufficient. In order to become self-sufficient, one needs to be allowed to pursue one's goals relatively unhindered. When markets aren't truly free but crippled by interventionists, self-sufficiency becomes harder to achieve. This skews the prices of stuff that people want in comparison with the prices of what they actually need. People who aren't even close to being self-sufficient but still have a steady income from social security programs buy frivolous things they don't really need all the time. The fact that you can be poor and obese at the same time is a testament to this. That combination was unthinkable everywhere in the world a mere hundred years ago. The entrepreneurs won't complain, though. Whether or not their customers are really acting in their own long-term best interest or not matters little to them. This is a feature, not a bug, in the free market system, and it does work, but only if the customer realizes that how they ought to allocate their resources is their problem and theirs alone. A semi-free market, distorted by interventionism, can never reach its full potential because of these effects.

Power to the People

Under a fiat currency monetary system, multinational corporations and big governments are hopelessly intertwined and inevitably help fuel each other's agendas. Policies such as safety regulations and standardizations may have been implemented by politicians with the best of intentions, but what inexorably happens when lobbyists advance their agenda is that laws and regulations set the bar for entry into the market at heights unreachable for smaller companies. Entire markets get monopolized by multinational behemoths because of centralized governance models. One example of this bureaucratic cancer is the General Data Protection Regulation (GDPR), the new set of rules governing how companies must store their employees' and clients' personal data in the EU. To comply, smaller companies will have to allocate a larger part of their total revenue than bigger companies, giving them a competitive disadvantage in the process. Giant media outlets will also likely benefit from Article 11 and Article 13, a copyright directive that will open the door to censorship of YouTubers and other content creators. The outdated copyright laws of the twentieth century are being shoehorned into the realm of the

Internet, where they were never supposed to be in the first place. Traditional copyright laws are obsolete and dysfunctional and will not protect the starving artist at all. The only thing they do is empower the big media houses and Hollywood studios.

You can vote for this or that party, but the truth is, as long as the control of the money supply remains centralized, there is nothing you can do to *really* influence the decisions of those with true power. All you can do is limit the ways in which they can control *you*. Bitcoin is a one-man revolution for every man, woman, and child on Earth. An individualist revolution is the most grassroots thing there is, but it does require people to actively participate rather than just complain about the state of things. *This is a good thing.* A lot of privacy tools are already available for those who want access to an escape tunnel now rather than later. Privacy-focused hardware such as Purism computers, privacy-focused web browsers, VPN tunnels, and the TOR project might all be somewhat flawed, but at least they're a step in the right direction. The revolution will not be centralized.

Bitcoin by itself is only pseudonymous at best, but combined with some of the aforementioned tools, a bunch of hardware wallets, and a full node, anyone can take a giant leap towards self-sovereignty for a relatively small amount of money. Remember that even if the Internet provides everyone on Earth with a plethora of opportunities as it is, if we want it to reach its full

potential, we need to keep it decentralized. This can only be done by actively choosing not to use the default settings on most gadgets and by being vigilant when prompted about what we want to allow our phones, tablets, and computers to do.

The idea of absolute digital scarcity achieved through decentralization is not easily dismissed, not even for politicians and the Silicon Valley elite. Some of the more honest among them embrace the idea, like Senator Patrick McHenry of the 10th district of North Carolina or Jack Dorsey, the CEO of Twitter. We'd better help these sorts of voices get heard if we want the mainstream to catch on. You don't *have* to be a virtue-signaling delusional psychopath to get somewhere in this world. There are more or less decentralized alternatives to Twitter, but as long as Jack's in charge, it certainly seems like one of the better social media platforms we have today.[1] True, all the big platforms have some form of censorship algorithm running, but remember that these networks are privately owned and entirely made up of people who voluntarily chose to join them. It's not really censorship if you have the choice to opt out. Censorship is only censorship when carried out by a government authority or another mafia-like entity because it is only then that you're coerced into

1. After this book was written, Jack Dorsey stepped down as CEO of Twitter and started devoting much of his time and energy to Bitcoin-related projects, such as Nostr.

silence. A voluntary system doesn't count, even though a more pro-free-speech attitude from the Facebooks and Googles of the world would probably be a good thing. Keep in mind how the Cantillon Effect helped these conglomerates amass power and that the ultimate tool for putting an end to their reign is in your hands already. Bitcoin. Fixes. This.

Every time you vote in a democratic election, you legitimize democracy. Most people in most democracies do this, even though *all* democracies are *all* flawed in more or less obvious ways. Every time you stack sats or acquire Bitcoin in whatever way you can, *especially* if you get paid in Bitcoin for a good or service, you promote yourself. By using Bitcoin, you vote for yourself and your personal independence. No matter what you think of the country you live in or democracy in general, I bet you wouldn't mind if the powers that be were a little less capable of patronizing *you*. We humans tend to think of ourselves as slightly more capable of deciding things for ourselves than our peers. With Bitcoin, we have a way of proving this to be true. Whether we're *actually* capable of handling our own private keys doesn't really matter to the system. Bitcoin will punish the reckless and compensate the attentive accordingly. It acts as the ultimate expression of equality of opportunity, and eventually, the Bitcoin economy will weed out those who weren't really mature enough to use it in the first place. This may sound harsh, but the truly, deeply evil thing that fiat currency does to people is arguably a lot worse.

Social benefits and safety nets have deprived people of the ability to take responsibility for the consequences of their actions and learn from their mistakes, creating a generation of snowflakes and unicorns who unscrupulously claim victimhood whenever anything bad happens to them. Everything is always someone else's fault, it seems. The saddest part about this development is that it takes the focus of the general public away from the real problems of the world and shifts it towards more and more arbitrary matters. If you feel that you've been bereft of your childhood, maybe you should try growing up.

Opting out of the current paradigm isn't easy, but it's easier than most people believe it to be. You don't *have* to choose one system or the other. Having just *some* Bitcoin is an insurance policy like no other because it hedges against *the entire system* collapsing. As the global number of Bitcoin users grows, individual nations will have a harder and harder time controlling it or even taxing its users. Just as the children of today dismiss any of the obsolete arguments about anything from their parents' generation with the phrase *"OK, boomer,"* so will the children of tomorrow dismiss concepts like "banking hours" or even "inflation." They will have known about Bitcoin all their lives, and they will have taken notice of its behavior over the years compared to that of the fiat currencies. With the discovery of Bitcoin, the adoption clock started ticking, and the inevitable fall of the fiat currency era began.

Slowly, Then All At Once

In 1869, Henry Heinz founded the Heinz Company, and a couple of years later, their now iconic (albeit slightly moronic) ketchup bottle was first introduced to the world. The bottle provides a perfect metaphor for how disruptive technologies spread over the globe. In the past, technologies were mainly mechanical and made up of physical stuff. This meant that producing them took time and, more importantly, that it took time for people to realize how their fellow man could improve his life by adopting the technology and thereby developing a personal demand for it. In other words, the process implied by *Metcalfe's Law* took a lot longer than it does now, in an era where technological progress is mostly computerized and made up of bits and bytes rather than physical building blocks. Metcalfe's Law states that the value of a communications network is proportional to its number of users, squared. For example, it took the automobile 62 years to reach 50 million users, while the same process took Facebook only 4 years. The credit card has been around since 1958 and it took 28 years for it to hit 50 million users. Improvements to the monetary system usually take longer for people to adopt than other technologies because they change societies at their very

foundation, especially if the technology alters existing power structures and hierarchies since those at the top of the food chain will always try to delay any development that threatens their advantageous positions.

When all the underlying technology needed for a new innovation to take flight is already in place, things can go very fast. The mobile game *Pokémon Go*, for instance, had been downloaded by 50 million users only 19 days after its release. All the infrastructure needed for the word to spread about the game was already there, in everyone's pocket. Bitcoin was first introduced to the world on January 3rd, 2009. It is impossible to know exactly how many Bitcoin users there are in the world because of Bitcoin's pseudonymous nature, but it's estimated that around 30 million people around the globe have at least some satoshis at the time of writing. Keep in mind that there's no limit to how many wallets a single person can have and one single wallet could act as a custodial service to multiple different users. It is also impossible to know whether old, seemingly inactive wallets are just made up of lost coins or if their owners are simply hodling and not spending their funds. Eleven years may seem like a long time for an Internet-based technology to spread, but remember that Bitcoin had no marketing behind it whatsoever. It took the mobile phone twelve years to reach 50 million users, and this technology had a multitude of multinational corporations behind it, all pouring vast amounts of resources into marketing their own product more aggressively than their competitors. Bitcoin grows organically, which means that it relies

solely on its users to educate their peers and spread the word about it. Some of the infrastructure needed for Bitcoin to flourish, like the Internet, was in place at its conception, but all of the yet-to-be-invented products, like hardware wallets, exchanges, steel tubes for private key protection, and Lightning ATMs, had to be invented by enthusiasts along the way. All of this while being constantly thwarted and opposed by those who stand to lose something from Bitcoin's existence, namely banks and governments.

The very meta, indeed inter-subjective, world of the in-game virtual goods market has an estimated value of around 50 billion dollars at the time of writing. Compare this to the total market cap of Bitcoin at the time of writing: 130 billion dollars. None of these items have any so-called "intrinsic value" or use case outside the virtual worlds they exist in. This market shows no sign of slowing down either. Just this very niche economy alone is a huge potential use case for Bitcoin in general and the lightning network specifically. Imagine a world in which gamers could stake and earn real Bitcoin for their in-game efforts. The gaming industry moves quickly, and once they realize what this means, those 50 billion dollars will be added to the Bitcoin economy in the blink of an eye. Who in their right mind would prefer lousy Warcraft gold or Linden dollars to real, absolutely scarce satoshis?

Hyperbitcoinization is a concept that is highly theoretical at this point in time, but as the Bitcoin

economy continues to grow, the likelihood that hyperbitcoinization will happen sooner or later increases exponentially. The exponentiality stems from Metcalfe's law primarily, and there are a bunch of domino effects that could trigger such an event. As the domino effects will fuel each other, hyperbitcoinization will probably play out a lot faster than its opposite, hyperinflation, normally does.

The first of the domino effects is the *Fear Of Missing Out*, or FOMO, from both people and financial institutions. When prices rise, people think they'll miss the opportunity if they don't hop onto the train fast enough, and a bull run ensues. This has happened in Bitcoin many times before, and each time it happens again, attentive market participants notice that this might be a trend rather than a bubble. Sooner or later, some central banker and maybe a finance minister here and there will notice this, too, despite their incompetence in these matters. In a very short period of time, it will be obvious that nation states which embrace this technology will get richer at a much higher pace than those who try to fight it. This will, in turn, trigger a much bigger FOMO cycle. A cycle in which nation states try to out-Bitcoin each other whilst at the same time trying to fight for their own survival, as the by then astronomical Bitcoin price has effectively turned every individual that was ahead of the game and got in early into a sovereign entity of the same financial magnitude as some of the nation-states.

Another domino effect that could trigger hyperbitcoinization is the currency wars currently being fought by the superpowers of the world: the United States, China, and the EU. Their virtual printing presses conjure up more and more dollars, yuan, and euros out of thin air, causing extreme short-term instability and, eventually, significant inflation. This has happened to every other fiat currency that ever existed and it's just a matter of time before it will happen to the ones in current use as well. Other central-bank-issued currencies might replace them, of course — the Euro is only twenty years old, after all — but every time this happens, the public will be a little more aware of the deceptive nature of fiat monies. There's only one alternative to inflationary currencies in the digital realm, and as long as it continues to do what it does, people will consider reallocating their wealth to it.

It's hard to imagine what a Bitcoin-denominated world would look like, but a natural effect of a more well-oiled free market and a more precise division of labor is that more and more people will be able to work less and earn more. It's easy to underestimate how powerful these mechanisms are because we've never truly seen them in action on a global scale before. All we have are historical examples of different eras and places where an almost-free market has been available and the enormous riches that this has brought to the people living there. Hong Kong is one such example.

An inflation-free global economy would make every good extremely cheap but would, at the same time, remove the need for merchants to store a surplus of everything as they do now, as everyone would be less prone to buy frivolous things they don't need because of the ever-increasing value of saving rather than spending. The velocity of money, which the currently dominant Keynesian economic theory holds as the most important metric in an economy, is only important if the market participants buy goods and services that make them more productive. In other words, it only works if people *invest* their money instead of wasting it on consumable goods. In a Bitcoin-denominated world economy, the total number of transactions per time unit would go *down*, while the number of unnecessary goods and services sold per time unit would go down *drastically*. In other words, the world would be a better, more effective, less stressful, and more environmentally friendly place.

Keep in mind that while it is true that Bitcoin is an experiment, the ever increasingly inflating, soon-to-be cashless, digital fiat monetary system is an even bigger experiment. Never before have the banks had such surveillance capabilities as they do now. All of your debit cards, all of your credit cards, and all of your banking apps are being monitored on a scale never seen before in history. Physical cash is being abolished around the globe at an ever-increasing rate. It's becoming easier and easier for everyone to spend more and more every day. Merchants are increasingly required to implement KYC and AML policies everywhere. Savings accounts have

negative interest rates, even if you don't take inflation into account. Some banks require their customers to spend a certain amount every month in order to keep their checking accounts. The list goes on and on. What this does, apart from building a nightmarish dystopian surveillance machine, is putting up immense barriers for everyone outside the system trying to get in. Good luck getting a decent loan if you're an ex-con, an undocumented refugee, a political activist, or even if you're just unemployed at the moment. Even if you believe that your soul belongs to some deity's supposed offspring, your ass belongs to the system as long as you stay dependent on it for your survival. So, how do we break the spell? How do we make our leaders change their minds and rethink the whole thing? It's simple — we don't. Because there is no "we". It's not them versus us. It's you versus everyone trying to control you. It has always been. Are you in the driver's seat or not? That's the question. Red pill or blue pill? Your choice.

Violence and Silence

Around the time that these words were written, a protest fund worth around 70 million dollars intended for the protesters of Hong Kong was frozen over "money laundering" by HSBC, acting under pressure from the Chinese government. *"Not your keys, not your Bitcoin"* is, despite what anyone might believe or wish for, just as true for the traditional banking system as it is for Bitcoin. Credit card companies, banks, and even Internet providers have immensely powerful tools for shutting people up at their disposal. Most of them are, fortunately, very reluctant to use these tools most of the time, but only because of the damage to their reputation a misdirected freezing of someone's funds could lead to. When not asked but ordered by a powerful enough authority to do so, most of them won't blink before ruining someone's life because of an alleged connection to "terrorism."

"Of all the numerous forms that governments have taken over the centuries, of all the concepts and institutions that have been tried, none has succeeded in keeping the State in check.", Murray Rothbard once concluded. Regardless of your opinion on the legitimacy of the state, it is undeniably very risky to grant any institution this kind

of power over people. You might like your government now, but if the "other side" wins the next election, you might not be too happy with a system that automatically grants them power over people's lives of this magnitude. The creepy thing about letting a third party decide what rules apply at what point in time and to whom is that the system might function perfectly fine until the day that it suddenly doesn't. You can't see it coming. That's the scariest part.

People in power very seldomly give up that power willingly. While this observation may seem obvious, it is less obvious what power truly is and where it truly resides. Voters seem to miss the target all the time and keep on voting for parties that suggest that giving politicians more power is a good idea. Regardless of where on the right/left spectrum the parties of most democracies today land, they all want themselves to have more power over other people. The right-wingers become more and more conservative rather than libertarian, more protectionist and nationalistic than pro-free trade, and so on. The left-wingers are, well, left-wingers, so they naturally want to increase taxes and control people's finances to a greater extent. Not to mention their self-harming tendency to focus on more and more cosmetic issues, like the political agenda-de-jour of the social justice movement. Deregulation, less governance, fewer taxes, free speech, and more personal responsibility are values that almost no politician rallies behind any longer. Furthermore, there's a tendency among the political class

and the old media houses to hide their pompousness by introducing vague terms to mislead the public, such as the "political compass," "filter bubbles," and various distortions of the term "common values." There's a strong connection between their not-so-subtle joint effort agenda, pushed by established media and politicians, and the recent rise in distrust of politicians in general. After the Internet came along and provided alternatives to whatever political opinions the establishment wanted to promote, political brainwashing machines have started to creak in their joints.

Hate speech is a term obviously coined by someone who hated speech. Nothing is as important to a society as free speech, and as discussed earlier, the only alternative humans have to speech when it comes to resolving conflict is violence. When lawmakers try to treat the symptoms of a societal problem with political band-aids such as hate speech laws, they not only fail to address the root cause of the problem, but undermine the very system that is protecting the victimized minority that the new law is supposedly designed to protect. Not only that, but they also cement the old group-think that labels people with categorizations that an unbiased society isn't supposed to take into consideration. *Diversity* is a term commonly used to describe everything except what truly needs to be diverse in a society, namely people's opinions. Instead, the word is used to describe cosmetic traits, such as skin color, gender, age, or sexual preferences. Governments all around the Western world brag about their "diverse" parliaments, focusing only

on the superficial whilst pushing a more and more unified political agenda. Anyone who dares to question if politicians and journalists are really qualified to conjure up solutions to supposed climate problems, for instance, is labeled a conspiracy theorist and anti-science. For anyone who has read any dystopian fiction in general, and Orwell in particular, the alarm bells should have sounded a long time ago.

Anyone who wants to be able to freely express his or her opinions in the future would be wise to act sooner rather than later. Your top priority should be to secure your financial freedom because without it, you're on a leash, and you have no reliable guarantees whatsoever. Money is a linguistic tool we use in order to express value to each other, and at this point in time, there is no nation on earth where people's ability to use money unhindered is protected by free speech legislation. Bitcoin, in particular, should be protected by free speech laws since Bitcoin is entirely made up of computer code and mathematics. If a law hinders you from expressing mathematics, you can instantly discard that law book as deeply immoral. Right now, this holds true for most law books in existence. This is not to say that anyone should actively break the law or not, just to emphasize that being moral is one thing and following the law another. Think hard and deep about where you stand on these issues. Chances are that there's nothing you can do about the society you were born into, but quite a lot you can do to increase your personal independence.

While you're at it, think about taxes. Think about who really pays them. Businesses need to be profitable in order to survive, so whenever a business is taxed, they will have to raise the prices of their products and services. Therefore, it is always the consumer that ultimately pays the tax. This is true for every tax there is.

VAT and sales taxes, for instance, are just a small part of the total amount of taxes that raise the price of a product, ultimately paid by the consumer. The employees of the company producing said product are probably paying income taxes, forcing their employers to pay them higher salaries. Employees also live in houses and have to pay real estate taxes, which ultimately forces the business owner to pay them more. Not to mention that the company itself needs to be physically located somewhere. A real estate tax is supposed to be a taxation of property owners, but in reality, it's always the consumer who pays the bill in the end. The tenant will have to pay rent, and if there's a real estate tax on the property, the owner will have to charge the tenant a higher monthly price.

All taxes ultimately punish the consumer, either through higher prices or a smaller supply of products to choose from. Prices can't be raised indefinitely, and some companies go out of business instead, lessening the competition between the remaining market actors and punishing the consumer even more. The taxes collected might be used for things a consumer would want anyway, like health care or education, but always through a process more ineffective and more expensive than that

which a truly free market would provide. In a truly free market with sound money, the division of labor mechanisms of that market would give us close to zero marginal costs for all products while at the same time disincentivizing people from overconsumption. True sustainability instead of mere buzzwords.

Mainstream Mediocrity

What is a Medium of Exchange? A monetary good is often described as a Store of Value, a Medium of Exchange, and a Unit of Account. But what do these descriptions actually mean, and in what order are they important for a monetary good to succeed, in the short and the long term? It all depends on the depth of one's analysis. Let's rewind to the dawn of civilized society. Money hasn't really been invented yet, and good old bartering is the only means of trading there is. "I will give you my three goats if you give me your cow." If the receiver of this proposal values three goats more than one cow, an exchange occurs. This is at the very core of all human interaction that isn't violent. Both parties believe that they stand to gain something from the interaction. If this wasn't the case, no interaction would have taken place. But how can I know that one or more of my three goats won't be used as barter in another exchange in which I'm not a participant? I can't, and neither can anyone else except the new owner of these bearded omnivores. They are, by definition, a *Medium of Exchange*. So is everything else. Every physical thing that anyone has ever claimed ownership of can be used as a

Medium of Exchange. That good's usefulness as such, however, is another matter.

A goat can hardly be considered a very effective form of value-bearing asset by anyone. In order for a good to be a useful Medium of Exchange, it not only needs to be portable, divisible, fungible, and not easily confiscatable, but it also needs to be able to store value, at least in the short term. Storing value is the trickiest part since what anyone finds valuable is entirely subjective. This is easily forgotten. The order in which the other monetary properties are important depends on how, where, why, and when the supposed exchange takes place. A car, for instance, can be considered relatively portable if it works and the potential buyer is within an acceptable driving range, but it's not very divisible and quite easy to confiscate. In-game gold or Monopoly money, on the other hand, is very divisible and portable but not very fungible since it's almost exclusively attractive to those playing a very specific game at a very specific point in time. Not only this, but in-game gold can be infinitely mined, and Monopoly money can easily be printed. The rules of Monopoly even say to just write numbers on a piece of paper when the bank runs out of bills! These examples might seem arbitrary and insignificant to any real-world economy, but the truth is that the only thing separating money from other goods as a Medium of Exchange is money's usefulness as such.

In order for anyone to accept something as payment for another thing, they need to be confident that this

something will not lose its value anytime soon. This is the one key property that any method of payment must hold. The greatest threat to this property is said medium's potential to be produced in large quantities over a short span of time. An apple can be bought for a certain amount of money, but no one would accept apples as payment for a new car since the apples would rot and lose their value well before they could exchange them for something else. What everyone seems to have forgotten is that the same is true for fiat money. Your dollars or euros won't rot overnight, but they will rot over a couple of decades. No one stacks cash in their mattresses anymore for this reason. Inflation deprives us of the ability to store the value of our labor long-term. Because of this, every decision made by every politician, every merchant, and every entrepreneur today is influenced by an unconscious incentive to focus on increasingly shorter-term benefits. Human progress is equipped with a damper, and we're not as progressive, effective, or innovative as we could have been. All because of our overlords' inability to resist the urge to dilute our money supply. The temptation to counterfeit has existed for as long as money has existed, and no civilization has ever been able to stop it. On the contrary, we've been very inventive in coming up with plausible excuses for printing money. The latest term for this criminal behavior is quantitative easing.

Enter Bitcoin — a monetary entity that no human can alter or even influence at this point in time. A very divisible token directly linked to the most fundamental

thing of value the universe has to offer — *energy*. A means of converting energy into a part of the world's only digital pie, of which no more than 21 million slices can ever be cut. A portable, divisible, fungible, and not easily confiscatable form of money that exists and proves its superiority to the existing system every day. So why isn't Bitcoin particularly popular as a Medium of Exchange? Will it ever be, and more importantly, does it really matter? To answer this, we must dive a little deeper into the subject. Bitcoin is a very good Store of Value from a personal perspective. You acquire an amount, and that amount stays the same over time. More importantly, that amount will represent the same part of the total sum of Bitcoin that can ever exist, no matter how long you choose to keep it. If you want a specific amount of Bitcoin to "buy you a Lambo," all you have to do is wait for someone to be willing to sell you a Lambo for that amount. This might take a while (or it might not), but if Bitcoin just manages to keep on doing what it does, it will happen. It is only a matter of time because, unlike Lambos, Bitcoins are scarce. Very scarce. Even absolutely scarce, which is a property of an asset that mankind has never encountered before. More and more people realize this every day, which is why they're reluctant to sell their Bitcoin. The newly minted Bitcoin that is mined every day needs to be sold in order for the miners' business models to work, but the Bitcoin that is already in circulation tends to stay where it is because people value the coins they possess a lot higher than what the current price in dollars or euros happens to be.

A good Medium of Exchange needs to be able to store value. The better it stores value, however, the less likely people are to exchange it for something that isn't likely to store value as efficiently. Bitcoin's value has such a large potential upside that people refuse to exchange it for frivolous things such as coffee or mass-produced cars. Contrary to what one might think, this doesn't make it a bad Medium of Exchange. Quite the opposite. How often a Medium of Exchange is used is not the correct metric to look at when trying to measure its usefulness as such. This misses the point. What should be measured is said medium's ability to buy you as much, or more, than what you bought it for. Bitcoin, because it is the only tool that cannot be diluted with new supply, is the closest thing possible to a guarantee of its long-term store of value.

Fiat money, inflation, and the ideas of John Maynard Keynes have distorted our perception of what money ought to be, so much so that most of us believe that coffee-buying convenience is the most important aspect of a monetary good. Looking only at merchant adoption and spread of acceptance metrics, Bitcoin still seems to be struggling quite a lot. In reality, these metrics are nonsequiturs and have very little to do with the actual success or functionality of the network. The price of a Bitcoin, as shown by ticker widgets and market cap websites, represents the lowest current price that anyone is willing to accept on an exchange market. Only a few Bitcoin owners are willing to accept this price, and most of them are waiting for a better opportunity. Bitcoin is indeed a currency, but it behaves very differently than all

other currencies that preceded it. The fact that people are reluctant to use Bitcoin for everything but a few very important transactions is a clue to Bitcoin's monetary superiority rather than a sign of anything else. So forget about coffee, forget about whether your local franchise burger joint accepts Bitcoin or not, and start focusing on what you can use this tool for.

Protection from What?

Some of the rules that governments choose to impose on us are supposedly there to protect us from ourselves and our respective shortcomings. Bank bailouts, for instance. By removing the punishment connected to risk-taking, you also inevitably remove the ability for people involved to learn from their mistakes. Having *skin in the game* is crucial for any learning process. If this is removed, the risk taker will learn the opposite of what he's supposed to learn from failure. He will conclude that the risk was worth taking, regardless of its consequences. Every time a law or policy mitigates the natural penalty for a bad investment, the entire system becomes a little more friendly to scammers and insincere people. This principle holds true for bank bailout policies but also for market regulations that primarily help the regulator and his friends by setting the bar for entry into the market at a height that their competitors can't overcome. However, the moral hazard created by removing people's skin in the game isn't limited to just financial markets and banks. Far from it. Safety regulations, standardizations, certifications, and even social security have successfully removed certain risks from certain professions and businesses, but in

doing so, they've also helped remove critical thinking and created entire species of bureaucrats that weren't needed in the first place. Bureaucrats, who have no skin in the game at all except for their own jobs, are conjuring up new policies as excuses for their existence all the time. While mitigating risk may be helpful in the short term, these policies are effectively turning what should be a concrete foundation to build our citadels on top of into a house of cards occasionally repaired with duct tape.

Proponents of generous social benefit policies often claim that their opponents should be less selfish and more empathetic to the needs of their fellow man. In truth, nothing makes people more selfish than a political system that continuously tells them that they're *entitled* to this and that. When people believe these lies, they also believe that they wouldn't have survived without the systems they were born into. They are made to believe that without the state and its coercive taxes, things simply wouldn't work, and no one would be there to look after them. The sense of entitlement is a very dangerous thing because it makes people believe that they can bypass personal responsibility from their life equation. It makes people forget that a society is just made up of its citizens and that there's no limitless faucet of wealth to pour new resources from whenever politicians see fit. Everyone needs to chip in in order for the machine to work. Fat beggars, able-bodied couch potatoes, and chain smokers demanding free healthcare are relatively new phenomena, and they're not proof of how far we've come or how humane our societies have become. On

the contrary, they're a testament to how limitless the cynicism of the elite really is and what lengths they're willing to go to in order to keep their voting cattle ignorant enough to uphold a decadent system that slowly deprives them of their souls.

Dependence is the poodle's kernel here. The more dependent the participants of a political system are, the less likely that system is to survive in the long run. As Margaret Thatcher said: "The problem with socialism is that you eventually run out of other people's money." Any society that makes its people dependent on the mechanisms of that society and demands nothing in return is doomed to fail. One could argue that social security programs make people less prone to criminal activity, but this argument fails to address the crime that was needed in order to anesthetize these would-be criminals in the first place, namely the coercive taxation methods used to finance the whole ordeal. Anything but personal responsibility for your actions is less effective and more costly for a society in the long run.

In Bitcoin, none of these problems exist, and it would be very hard to implement such policies in a Bitcoin economy because of the very nature of Bitcoin's cryptographic signature model. Not your keys, not your Bitcoin is just another way of saying no pain, no gain, and as everyone who has ever truly owned and lost Bitcoin knows, this is the true nature of any possession. If your possessions are somehow guaranteed protection by a third party and not by yourself, you don't truly own

them. With Bitcoin as a world reserve currency, you can build a society on top of an underlying concrete trustless foundation, and you can weave this proveable sincerity into the fabric of whatever policies and values society aims to uphold. It could provide us with a much more stable common ground than fictitious promises such as the UN's list of basic human rights, the US Declaration of Independence, or even the Ten Commandments ever could. The only thing preventing this from happening is our collective lack of imagination.

The One Shot Principle and The Always Up Point

Who am I to conjure up theories about macroeconomics? I have no financial background, and even though I've probably studied mathematics more than your average Joe, that was a long time ago, and I'm hardly an expert at anything, really. Well, if there's one thing that Bitcoin has taught me, it's how scarce my time is and that I won't be able to excel at anything if I don't have skin in the game. Everyone is new to Bitcoin, and everything in economics is speculative. John Cleese, one of my favorite human beings of all time, has said that it seemed like the main goal in life for everyone in his hometown of Weston-super-Mare was "to get safely into their coffins without ever having been seriously embarrassed." This phrase struck a chord with me. After having discovered Bitcoin and, in doing so also having discovered new ways to think about scarcity and time, I've decided to not hold my brain back any longer but to put my thoughts out there, where they at least might ignite a spark in someone else's mind. Standing on the shoulders of true giants, here's my take on an economic theory and an explanation of the specific event that it implies.

The One Shot Principle:

> *Absolute mathematical scarcity achieved by consensus in a sufficiently decentralized distributed network was a discovery rather than an invention. It cannot be achieved again by a network made up of participants aware of this discovery since the very thing discovered was resistance to replicability itself.*

Absolute scarcity on the Internet was long believed to be impossible because of the very nature of data. Ones and zeros that could be copied an infinite number of times weren't believed to be capable of representing something as uncompromisingly finite as Bitcoin's 21 million coin issuance cap. That is until Satoshi Nakamoto showed that the double spending problem could be solved by giving every participant in the network a reward for following the rules and making sure that every other participant also did so. In doing this, and then disappearing at an early stage of the network's development, he gave enough participants hope that the decentralization experiment could actually work and created a living thing that feeds on human incentives.

The rise of Bitcoin was a true black swan event, and it cannot happen again. A competing token would need to be sufficiently different from Bitcoin in order to achieve true decentralization because it would be trivial for any big player to completely dominate this

new coin, and there would be no way to introduce it to the market without a skewed initial distribution. A competitor would need to differentiate itself so much from Bitcoin that when push came to shove, it wouldn't be a competitor at all.

More importantly, if a "technically superior" or "more efficient" token one day renders Bitcoin a thing of the past, the whole concept of absolute scarcity on the Internet can be considered a failure because there would be no guarantees whatsoever that this new token wouldn't suffer a similar fate at some point in the future. Users would no longer stand to gain anything from saving their Bitcoin, but contrary to this, they would be incentivized to spend and acquire new, alternative tokens. This would destroy the only true value proposition of Bitcoin, its scarcity, and deprive any future attempt at creating decentralized currencies of any legitimacy. This is the main reason that blockchains other than Bitcoin's blockchain are counterproductive and confusing to a lot of newcomers, especially since a lot of Bitcoin proponents don't seem to understand this fully themselves.

Hal Finney, the man who received the first ever Bitcoin transaction from Satoshi Nakamoto, definitely understood, though. He said the following in May 2011:

> "Any successful replacement of the Bitcoin blockchain will forever undermine the credibility of any successor. How can an

> investor know that it won't happen again? Rebooting now may benefit a few thousand early adopters. What happens when hundreds of millions use Bitcoin 2.0? They'll be just as jealous and envious of you as you are of others. Given the precedent you want to set, how will you argue against yet another reboot?"

For those of you who don't know, Hal was one of the people closest to Satoshi and a monumental part of Bitcoin's early history. He believed that "the computer could be used as a tool to liberate and protect people, rather than to control them," and some people even believe that he was the man behind the pseudonym. This would explain why Satoshi went silent in 2011 since Hal fully lost control of his locomotive functions around the time of Satoshi's disappearance because of the Amyotrophic Lateral Sclerosis disease he suffered from. The disease eventually bereft him of his life on August 28, 2014. Hal's body is cryopreserved and poised for resurrection as soon as this has been invented. Any day now...

The Always Up Point:

The Always Up Point is a highly theoretical, albeit logical, conclusion of what one deflationary currency competing with nothing but inflationary currencies might ultimately lead to. The Bitcoin block subsidy, which is the newly-minted-coins part of the block

reward, is halved every four years, resulting in an ever-increasing *stock-to-flow ratio*[1] of the asset. At some yet unknown point in the future, Bitcoin's unmatched stock-to-flow ratio combined with enough people who aren't willing to ever sell their Bitcoin, also known as *hodlers of last resort*, will make Bitcoin so hard to come by that it can't lose value. In other words, a point in time from where the price of a Bitcoin can only go up. This might sound outlandish, but keep in mind how far off in the future such a point may be and that all other currencies on Earth are inflationary, including gold. The comparison to other currencies is sort of misleading in this case since Bitcoin's monetary properties are so vastly different from what we've seen in the past.

Absolute scarcity means that one day there will only be 0.00000001 Bitcoin (or one *satoshi*) left in circulation. This satoshi will probably be worth more than an original painting by Leonardo Da Vinci, of which there are about thirteen left today at that point in time. It will simply be immeasurably valuable as a collectible. Not for sale at any price. The Always Up Point would theoretically happen a long time before this. Sometime between now and when all of the Bitcoin is lost, there's a tipping point where it becomes pointless to talk about Bitcoin's price in dollars, euros, or yuan, but the Always Up Point would precede that event, too. These two points may be closer to each other than they are to the present, or the point where

1. This concept is explained in the Laws and Effects chapter at the end of this book.

there is no Bitcoin left, but no one can know for sure how far off, or close, these two points really are.

The Always Up Point may sound like total science fiction, but try to think about these things without coming to these conclusions eventually. I bet you can't. There is, of course, always the possibility that Bitcoin may fail for one reason or another, but if it doesn't fail, these events are bound to happen and are not fictional at all.

The Secular Individual

According to Wikipedia, a *religion* is a "social-cultural system of designated behaviors and practices, morals, worldviews, texts, sanctified places, prophecies, ethics, or organizations, that relates humanity to supernatural, transcendental, or spiritual elements." However, there is no scholarly consensus over what precisely constitutes a religion. One could say that religions are inter-subjective sets of unverifiable beliefs that large groups of individuals hold at the same time.

Similarly, a *cult* is defined as a "social group that is defined by its unusual religious, spiritual, or philosophical beliefs, or by its common interest in a particular personality, object or goal." There are many different opinions about what distinguishes a cult from other groups of believers in inter-subjective things. In one of the examples of a definition on Wikipedia, a cult is defined as a social movement that has the following characteristics:

1. Authoritarian ruler. The self-appointed rulers of cults have complete and final say. They are usually considered charismatic and charming.

2. The identity of the community becomes communal and totalistic. Everyone involved relies on each other and the group's wants and needs become the core identity for each follower.

3. Aggressive campaigns and conversion efforts are enforced by the authority figures. For most, the fear of heaven and hell prompts members to reach out, but authority figures may also rely on social issues.

4. In order to be initiated into the group, there will be enforced and systematic indoctrination. There are usually several practices or ceremonies new followers must complete to secure their place among the group.

5. Religious movements labeled cults are usually quite new and do not have the established title of other religious practices. They also tend to build upon old theology and either update it to modern times or adapt it to their teachings.

Now compare these characteristics to any modern democratic nation-state and see if you can find one that cannot be described as a cult. Authoritarian ruler? *Check*. Communal and totalistic? *Check*. Aggressive campaigns? *Check*. Systematic indoctrination? *Mega-check*. In fact, the only bullet point whose applicability to the nation-state could be somewhat questioned here is point number five, the one about religion and old theology. *But not really*. Ceremonial burial, holidays on traditionally

"sacred" days, no work on Sundays, marriage, and name-giving ceremonies are all very old theological practices shoehorned into our current systems in order to not upset the established order too much. Not to mention the "In God we trust" banner printed on every US dollar bill in existence.

Secularity is the state of being separate from religion or of not being exclusively allied with or against any particular religion. The idea that nation-states ought to be secular and that there should be a distinct dichotomy between what the state should consider to be religious belief or not was born in Europe during the Enlightenment. Secularity on a national level has put an end to many barbaric religious rituals of the past and saved a great number of innocent victims from the repercussions of these practices. Forced marriage, female genital mutilation, and even stoning still exist in some parts of the world, but they are considered very cruel and primitive by the vast majority of people in states that try to uphold Enlightenment values.

While the separation of church and state was all nice and dandy, it is hard to see it as something other than a small first step towards something even greater once you've accepted the idea that there's really no clear distinction between religious and political dogma. Democratic countries still engage in warfare, and they still imprison people for victimless crimes, such as owning a gun or a specific plant. We're not immune to barbarism until we're truly self-sovereign. In order for

society to be truly secular, we need to separate not only church and state but money and state. In other words, we have to separate ourselves from the state or at least minimize its influence over our lives. This will never be done by the state itself, but requires individuals to take matters into their own hands. Those who don't aren't ever really secular but shackled to whatever set of beliefs their nation's leaders hold. Politics has never truly been about "us" versus "them". It has always been, and still is, about you versus those who wish to control you.

If a mobster with a gun knocks on your door and demands payment for your "protection," the wisest response might be to pay him off. It may also be wise not to slander or talk about him and his organization behind his back. That doesn't mean you have to agree with his worldview or find his behavior to not be morally reprehensible. The thoughts in your head are always your last bastion of freedom. No matter the threat, you're always free to do as you please inside the dome of your skull. Remember this because one day, the truth will truly set you free if you need it to. The truth of twenty-four words, memorized. The act of memorizing the seed of a Bitcoin wallet has already saved the lives of an unknown number of refugees, and one day, you might need to carry your twenty-four words in your head, too. The more of your wealth you store in the physical realm, the easier it is to confiscate what you have. Twenty-four words in a brain, on the other hand, can't even be detected. It is impossible for a perpetrator to know that you even have them. The so-called brain wallet is a

very underrated concept and one of the most profound aspects of what Bitcoin can do for you. It completely redraws the political world map for those who know about it, blurring out the border lines between nations and turning otherwise dangerous places into potential markets. When the money already exists everywhere, any limit on what amount you may carry from one nation to the other is as laughable as if there was a limit on how many languages you are allowed to know. The impact this will have on the financial freedom of individuals and, in turn, the world economy is vastly underestimated. The practice is still unknown to most people but real nonetheless. Like with everything else in Bitcoin, it is only a matter of time before people begin to realize what's going on.

Clear your mind. Start from scratch. Unlearn what you have learned. Every educational institution you've ever been to exists solely because of someone's agenda. If you were a straight "A" -student, you excelled at being average. Congratulations, you're a sheep. The core science subjects are true because the experiments can be verified. The social sciences are all biased and tailored for a certain narrative. You're born, you live, and you will most probably die eventually. The laws of nature govern everything in between. All societies are just inter-subjective constructs of the human mind. Choose your life's focus carefully. If you don't, you might trifle away your existence, believing fairy tales to be true.

Laws and Effects

This chapter is dedicated to the concepts, natural laws and effects that explain why Bitcoin is so much more than the sum of its parts. Understanding these concepts is crucial when defending Bitcoin during conversations with intelligent precoiners. Many of these concepts, if not all, have been discussed earlier in this book or in my first book, *Bitcoin: Sovereignty through Mathematics*, but this chapter is intended as a guide to wrapping your head around and being able to explain these concepts to newcomers with more ease. Scratching the surface, if you will. Bitcoin is hard to understand in general, but to really comprehend why it may well be the most important technological revolution you'll encounter in your lifetime, you'll need to understand the *why* before even trying to grok the *how*.

Subjective Theory of Value

Carl Menger, the founder of the Austrian school of economics, described the Subjective Theory of Value as "the idea that the value of a good is not determined by any inherent property of that good, nor by the amount of

labor necessary to produce the good, but instead by the importance an acting individual places on a good for the achievement of his desired ends." In other words, what anyone finds valuable is an entirely subjective matter. Valuability is always personal and situational, and there is little reason to question this insight. Keeping this fact in mind is crucial to understanding why no economic theory can ever be used to predict anything with perfect accuracy. Every investment is a guessing game and requires the investor to take as many factors (potential risks and rewards) as possible into consideration before making the trade.

Supply and Demand

Holding all else equal, the unit price for a particular good or service will vary until it settles at a point where the quantity demanded will equal the quantity supplied, resulting in an economic equilibrium of price and quantity transacted. Even though demand (which could also be called *inter-subjectively assigned value*) is subjective, its relationship with the supply of a certain good or service has an obvious connection to the price of said good or service. The more abundant the supply, the lower the price, and vice versa. It is important to remember that this theory can be applied to money just as much as it can be applied to any other good or service.

Stock-to-Flow

The impact an increase in the available supply of a good will have on its price is correlated to the size of the existing *stock* of that good, compared to the increase per time unit, or *flow*. This is called the *stock-to-flow ratio*. If increasing the supply of a good is difficult and, therefore, expensive, the impact an increase in the price of that good will have on its subsequent total supply is limited. Therefore, the increase will not push the price down again as much as it would have if increasing the supply would have been easy or cheap. Remember that the supply of a good is directly correlated to its price if the demand for it doesn't change. The high stock-to-flow ratio of precious metals is what makes them expensive. The fact that you can make jewelry out of some of them doesn't affect the price in the same way at all. It's the other way around. Expensive jewelry is made out of expensive materials because jewelry is a form of Proof-of-Work (described below) or, at the very least, a proof of wealth.

Gresham's and Thier's Laws

Gresham's Law states that "if there are two forms of commodity money in circulation, which are accepted by law as having similar face value, the more valuable commodity will gradually disappear from circulation." If we consider Bitcoin a commodity, it would gradually disappear since its absolute scarcity would render

Bitcoin the most valuable commodity in the world over time. Keep in mind that even if this happens, and Bitcoin isn't ever used as "money" ever again, its value can still go up indefinitely. There is, however, a sort of counter-theory to Gresham's Law. *Thier's Law* shows that in the absence of effective legal tender laws, Gresham's Law works in reverse. If given the choice of what money to accept, people will transact with the type of money they believe to be of the highest long-term value. However, if not given the choice and required to accept all money, good and bad, they will tend to keep the money of greater perceived value in their possession and pass on the bad money to someone else.

The Cantillon Effect & WTF Happened in 1971

As described earlier, Richard Cantillon noticed that original recipients of new money enjoy higher standards of living at the expense of later recipients. The concept of relative inflation (a disproportionate rise in prices among different goods in an economy) is known as the Cantillon Effect. Inflation is the main mechanism that funnels wealth into the pockets of the elite and away from everyone else. The further you are from the new money created, the more you pay. When President Nixon practically legalized counterfeiting in 1971 by abandoning the gold standard entirely, he gave the US the ability to literally trade worthless green paper for real commodities, and everyone on earth is still falling for it today. Many citizens of the United States became very

rich in the process because of the Cantillon Effect and were thus less likely to turn against this monetary policy.

Zero Marginal Cost

As Jeremy Rifkin argues in his book named after the phenomenon, most goods and services in the world are approaching a *zero marginal cost*. Intense global competition forces entrepreneurs to introduce ever more efficient technologies, accelerating productivity to the point where the marginal cost of production approaches zero, making goods and services almost free. We're in an era in which consumers are transforming into prosumers. Anyone can create music, art, videos, and books and distribute them to the world via the Internet for almost no cost at all. This book (and its predecessor) are great examples of this. They're printed only whenever so demanded by a new customer and marketed solely by social media. I didn't spend a single penny creating these books except for what I paid for the electricity and the Internet connection. In a world that has been so optimized by global capitalism that everything is nearly free to produce, all prices ought to reflect this phenomenon. Still, prices go up. The only thing hindering us from reaping the true fruits of our massive global collaborative efforts is our lack of sound, non-inflationary money. In a sound money economy, prices would go down, not up, over time.

The First Law of Thermodynamics

The *First Law of Thermodynamics*, also known as the *Law of Conservation of Energy*, states that energy cannot be created or destroyed in an isolated system. Bitcoin can express how much energy that was sacrificed in order to acquire a share of a limited supply. You can, of course, also acquire Bitcoin by buying it rather than mining it, but in doing so, you also spend energy. You somehow acquired the money you bought the Bitcoin with, and that somehow came to be because someone sacrificed time and energy somewhere else. Bitcoin lets you express that you see that there's a connection between value and scarcity by sacrificing effort or labor to be a part of the network. The value of the total amount of sacrificed labor cannot leave the isolated system. Therefore, theoretically, the total value of the network cannot decrease over time, even if the price does.

The Price of Money and the Prices of Everything Else

This is a sort of embryo of a theory I came up with while writing this book. It's just a thought at this point in time but well worth mentioning while explaining money in general and Bitcoin in particular to someone else. The theory goes something like this:

> "The lower the cost of money production, the higher the prices of everything else and vice versa."

In today's society, money is almost the cheapest commodity to produce in the world. Phenomena such as negative interest rates are a testament to this. The only things that become cheaper over time in an ever increasingly inflationary economy are those that become cheaper to produce at a rate faster than that of the monetary debasement. If money, on the other hand, was the most expensive commodity to produce, everything else would be cheaper in comparison to it. More for less. In a sound money economy, the market economy would function several orders of magnitude better than it does now because it would disincentivize frivolous spending and promote long-term investment instead.

Metcalfe's Law

Metcalfe's Law states that the effect of a telecommunications network (and therefore also its value) is proportional to the square of the number of connected users of the system. A single telephone is useless, but the value of every telephone increases exponentially with the total number of telephones in the network because the total number of people with whom each user can call and receive calls increases. Imagine this Law's impact on Bitcoin's value, where expressing value is the sole purpose of the network, and where this

is achieved through dividing a fixed number among the users. To say that the thought is mind-blowing would be an understatement.

The Byzantine General's Problem

The Byzantine General's Problem describes a condition of a computer system, distributed computing systems in particular, where components may fail, and there is imperfect information on whether a component has failed. Imagine that several divisions of the Byzantine army are camped outside an enemy city, each division commanded by its own general. The generals can communicate with one another only by messenger. After observing the enemy, they must decide upon a common plan of action. A traitor among the ranks could deliver false information and prevent the group from reaching consensus. The generals must, therefore, develop an algorithm to guarantee that all loyal generals decide upon the same plan of action and that a small number of traitors cannot cause the loyal generals to adopt a bad plan. Now, think of a traitor as a malicious party within a distributed ledger that aims to facilitate fraudulent transactions. In accordance with Metcalfe's Law, as the number of parties in the system increases, the number of channels for communication increases exponentially. Each communication channel could be used to carry false information. Imagine the complexity of building consensus in a truly decentralized system with thousands or millions of parties involved. This is

the main computer science problem that Bitcoin solved by aligning participants' incentives in such a way that cheating becomes very risky and expensive.

The Black Swan Theory

According to Wikipedia, a *Black Swan Event* is "an event that comes as a surprise, has a major effect, and is often inappropriately rationalized after the fact with the benefit of hindsight." The term was popularized by Nassim Nicholas Taleb. The discovery of Bitcoin was such an event. Almost no one believed that a currency native to the Internet could ever be created due to the nature of data itself and its infinite replicability. We're still in Bitcoin's infancy, and most people are still reluctant to believe that the network will work long-term.

The Lindy Effect

The Lindy Effect is a theory that the future life expectancy of some non-perishable things like a technology or an idea is proportional to their current age so that every additional period of survival implies a longer remaining life expectancy. The longer Bitcoin survives, the higher the probability that it will survive in the long term.

Roko's Honey Badger

The concept of *Roko's Basilisk* has been called the most dangerous thought ever imagined. It describes a non-existent artificial intelligence punishing everyone who didn't contribute to bringing about its eventual existence retroactively. Rational people would, therefore, help it come about just out of fear of what could potentially happen to them if they didn't, similar to why some people believe in God because of the supposed rationality described by *Pascal's Wager*. Bitcoin could be said to be a sort of inverted Roko's Basilisk in a way, since not having any Bitcoin if (when) hyperbitcoinization happens will be at least indirectly punished as such an event would render those that do have Bitcoin richer than those who don't at a very fast pace. This only holds true if you think that relative wealth gaps are inherently important though. As described in earlier chapters, wealth gaps are of little importance to Austrian economics. What's important is if you're better off than you have been historically, no matter what anyone else owns or earns. If you rethink this supposedly rationalistic phenomenon you end up with *Roko's Honey Badger* instead — an entity that rewards those who help it come alive rather than punishing those who don't.

Mass Adoption and Exponential Growth

So, where are we now?

At the time of writing, it is estimated that around 1% of the world's population own some Bitcoin. During the first eleven years of Bitcoin's existence, the number of Bitcoin users worldwide has doubled every year and even quadrupled during bull market years. As Albert Allen Bartlett said: "The greatest shortcoming of the human race is our inability to understand the exponential function." At the same adoption rate as that which we have now, not taking bull runs into account, more than half the planet's population will own Bitcoin in less than seven years. 2% in the first year from now, 4% in the next, then 8%, 16%, 32%, 64%, and so on.

So cut out the noise, buckle up, and enjoy the ride. Bitcoin changes everything. Ignore it at your own peril.

About the Author

Knut Svanholm is a Bitcoin author, educator, armchair philosopher, international village idiot, and rock star. His first work, '*Bitcoin: Sovereignty through Mathematics*,' is widely regarded as a classic in the space, earning recognition as a recommended read on numerous educational platforms, including MicroStrategy's Bitcoin site hope.com.

However, Knut's influence extends beyond his written contributions; he is credited with coining the popular meme 'Everything Divided by 21 Million,' a phrase that encapsulates the scarcity and mathematical principles underpinning Bitcoin. Armed with various superpowers, such as an ability to channel the power of dad jokes and going into "rant mode" at will, Knut dedicates his time to advising a select group of Bitcoin companies and hosting 'The Bitcoin Infinity Show,' a thought-provoking podcast exploring Bitcoin philosophy and its implications for individual sovereignty.

Knut's other works include:
Bitcoin: Sovereignty Through Mathematics (2019)
Bitcoin: Everything Divided by 21 Million (2022)
Praxeology – The Invisible Hand that Feeds You (2023)
Bitcoin: The Inverse of Clown World (with Luke de Wolf, 2024)

